which predictions for measurement outcomes could be extracted: for any particular outcome, the wave function assigns an amplitude, and the probability of getting that outcome is the amplitude squared. Max Born, who proposed the Born rule, was one of the attendees at Solvay in 1927.

Now we're saying something deeper and more direct. The wave function isn't a bookkeeping device; it's an exact representation of the quantum system, just as a set of positions and velocities would be a representation of a classical system. The world *is* a wave function, nothing more nor less. We can use the phrase "quantum state" as a synonym for "wave function," in direct parallel with calling a set of positions and velocities a "classical state."

This is a dramatic claim about the nature of reality. In ordinary conversation, even among grizzled veterans of quantum physics, people are always talking about concepts like "the position of the electron." But this wave-function-is-everything view implies that such talk is wrongheaded in an important way. There is no such thing as "the position of the electron." There is only the electron's wave function. Quantum mechanics implies a profound distinction between "what we can observe" and "what there really is." Our observations aren't revealing pre-existing facts of which we were previously ignorant; at best, they reveal a tiny slice of a much bigger, fundamentally elusive reality.

Consider an idea you will often hear: "Atoms are mostly empty space." Utterly wrong, according to the AQM way of thinking. It comes from a stubborn insistence on thinking of an electron as a tiny classical dot zipping around inside of the wave function, rather than the electron actually *being* the wave function. In AQM, there's nothing zipping around; there is only the quantum state. Atoms aren't mostly empty space; they are described by wave functions that stretch throughout the extent of the atom.

The way to break out of our classical intuition is to truly abandon the idea that the electron has some particular location. An electron is in a *superposition* of every possible position we could see it in, and it doesn't snap into any one specific location until we actually observe it to be there. "Superposition" is the word physicists use to emphasize that the electron exists in a combination of all positions, with a particular amplitude for each one. Quantum reality is a wave function; classical positions and velocities are merely what we are able to observe when we probe that wave function.